AI-Enabled University Queueing Analysis

Contents

SECTION 01

Executive Summary

This engineering analysis compares a legacy service desk with an AI-enabled operating model. All values are illustrative and are used only to verify professional mathematical rendering.

SECTION 02

Derivation of Mean Waiting Time (W_q)

The queueing model uses the Erlang-C probability of waiting and the service-capacity margin.

$$\text{Mean waiting time} \quad W_q = \frac{P(W > 0)}{c\mu - \lambda}$$

SECTION 03

1. Legacy System Erlang-C Calculation (c_0 = 15, ρ_0 = 0.80)

The legacy scenario uses fifteen parallel service channels at eighty percent utilisation.

$$P_0 = \left[\sum_{n=0}^{c-1} \frac{a^n}{n!} + \frac{a^c}{c!\,(1-\rho)} \right]^{-1}$$

SECTION 04

2. AI-Enabled System Calculation (c_1 = 8, λ_1 = 36, μ_1 = 25, ρ_1 = 0.18)

The AI-enabled scenario has lower utilisation and a larger service-capacity margin.

$$c_1 = 8, \quad \lambda_1 = 36, \quad \mu_1 = 25, \quad \rho_1 = 0.18$$

SECTION 05

Mathematical Result Summary

The modelled reduction is reported with its unit and without unsupported extrapolation.

$$\Delta W_q = W_{q,0} - W_{q,1} = 12.4 \text{ minutes}$$

$$\text{Total annual benefit} = 630000 + 120000 = 750000$$

SECTION 06

Conclusion

The labelled equations, Greek symbols, subscripts, units, and derivation notation remain readable while ordinary prose stays in paragraph form.